

CSS3 – Module 1 Answers

Local Community Event Portal | Exercises 1 – 11

Exercise 1: Why CSS? Inline vs Internal vs External

We apply styles in all three ways in the same HTML file, then move reusable rules to an external styles.css.

index.html

```
<!DOCTYPE html>
<html lang='en'>
<head>
<meta charset='UTF-8'>
<title>Local Community Event Portal</title>

<!-- Internal CSS (embedded in <head>) -->
<style>
/* Body background - internal style */
body {
background-color: #f0f4f8;
font-family: Arial, sans-serif;
margin: 0;
padding: 0;
}
</style>

<!-- External stylesheet -->
<link rel='stylesheet' href='styles.css'>
</head>
<body>

<!-- Inline style makes this heading red -->
<h1 id='mainHeader' style='color: red;'>
Welcome to the Community Event Portal
</h1>

<h2>Upcoming Events</h2>
<div class='eventCard'>
<h3>Summer Fair</h3>
<p>Join us for food, fun, and community!</p>
</div>

</body>
</html>
```

styles.css (external stylesheet – reusable rules)

```
/* =====
External Stylesheet - Community Event Portal
===== */

/* Header styles */
#mainHeader {
font-size: 2rem;
padding: 20px;
}
```

```
/* Event card container */
.eventCard {
background-color: #ffffff;
border: 1px solid #ccc;
border-radius: 8px;
padding: 16px;
margin: 12px 0;
}
```

[Screenshot – Browser with all three CSS methods applied]

Browser renders:

- Page background: light blue-grey (#f0f4f8) [internal CSS]
- H1 text colour: red [inline style]
- .eventCard has white bg + border [external CSS]

DevTools > Sources > styles.css is listed under Loaded Resources.

DevTools > Elements > h1 shows: style='color: red;' (inline, highest priority).

Note: Priority order: inline > internal/external (source order breaks ties). Move all rules to external CSS in production to keep HTML clean.

Exercise 2: CSS Syntax and Comments

Add structured, well-commented sections in styles.css. Consistent indentation (2 spaces) and a comment above every selector.

styles.css – formatted with comments

```
/* =====  
  
Reset  
===== */  
  
* {  
margin: 0;  
padding: 0;  
box-sizing: border-box;  
}  
  
/* =====  
  
Body  
===== */  
  
body {  
background-color: #f0f4f8;  
font-family: 'Segoe UI', Arial, sans-serif;  
color: #333333;  
line-height: 1.6;  
}  
  
/* Style for main CTA button */  
.cta-button {  
background-color: #007BFF;  
color: white;  
padding: 10px 20px;  
border: none;  
border-radius: 4px;  
cursor: pointer;  
}  
  
/* Hover state for CTA button */  
.cta-button:hover {  
background-color: #0056b3;  
}  
  
/* Event card container */  
.eventCard {  
background-color: #ffffff;  
border: 1px solid #dddddd;  
border-radius: 8px;  
padding: 16px;  
margin: 12px 0;  
}
```

[Screenshot – styles.css open in VS Code with comments visible]

VS Code: all rules indented 2 spaces, every selector has a comment above it.
Browser: no visual change from comments – they are stripped by the parser.
DevTools > Sources > styles.css shows clean, readable rules.

Exercise 3: Selectors Playground

Use all five required selector types in styles.css.

styles.css – all selector types

```
/* Universal selector - reset margin and padding on everything */
* {
margin: 0;
padding: 0;
box-sizing: border-box;
}

/* Element selector - style every h2 on the page */
h2 {
font-size: 1.5rem;
color: #2c3e50;
margin-bottom: 8px;
}

/* ID selector - the main banner heading */
#mainHeader {
background-color: #2c3e50;
color: white;
padding: 24px 16px;
text-align: center;
font-size: 2rem;
}

/* Class selector - every event card */
.eventCard {
background: white;
border: 1px solid #ccc;
border-radius: 8px;
padding: 16px;
margin: 12px 0;
}

/* Grouping selector - h3 and p share these rules */
h3, p {
font-family: 'Segoe UI', sans-serif;
color: #444;
margin-bottom: 6px;
}
```

[Screenshot – DevTools Elements panel showing applied selectors]

```
Browser - DevTools > Elements:
* rule applied: margin=0, padding=0 on <body>, <div>, etc.
h2: font-size 1.5rem, color #2c3e50
#mainHeader: dark background, white text, centred
.eventCard: white card with border and border-radius
h3, p: same font-family and colour applied to both elements
```

Exercise 4: Color and Background Styling

Use HEX, RGBA, a background image with fallback, and a linear gradient on section headers.

styles.css – colours and backgrounds

```
/* Body: background image with HEX fallback colour */
body {
background-color: #f0f4f8; /* fallback */
background-image: url('images/bg-pattern.png');
background-repeat: repeat;
background-size: auto;
color: #333333; /* HEX text colour */
}

/* Section header with linear gradient */
.section-header {
background: linear-gradient(135deg, #1a73e8, #0d47a1);
color: white;
padding: 14px 20px;
border-radius: 6px 6px 0 0;
font-size: 1.2rem;
}

/* RGBA – semi-transparent overlay on event cards */
.eventCard {
background-color: rgba(255, 255, 255, 0.92);
border: 1px solid rgba(0, 0, 0, 0.1);
padding: 16px;
margin: 12px 0;
border-radius: 8px;
}

/* Accent text colour using HEX */
.event-date {
color: #e74c3c;
font-weight: bold;
}
```

[Screenshot – Browser showing gradient header and RGBA cards]

Browser screenshot shows:

- Page background: pattern image tiles over #f0f4f8 fallback
- .section-header: blue gradient from #1a73e8 to #0d47a1
- .eventCard: slightly transparent white (rgba 0.92 alpha)
- .event-date text: red (#e74c3c)

If bg-pattern.png is missing, page still shows the #f0f4f8 colour.
(Tested by removing the file – fallback confirmed working.)

Exercise 5: Typography – Fonts and Text

Import a Google Font and apply text properties across the page.

index.html – Google Font import

```
<head>
<!-- Method 1: <link> tag (preferred) -->
<link rel='preconnect' href='https://fonts.googleapis.com'>
<link rel='preconnect' href='https://fonts.gstatic.com' crossorigin>
<link href='https://fonts.googleapis.com/css2?
family=Roboto:wght@400;700&family=Playfair+Display&display=swap'
rel='stylesheet'>
<link rel='stylesheet' href='styles.css'>
</head>
```

styles.css – typography rules

```
/* Base body font */
body {
font-family: 'Roboto', Arial, sans-serif;
font-size: 16px;
font-weight: 400;
line-height: 1.7;
color: #333;
}

/* Page main heading */
#mainHeader {
font-family: 'Playfair Display', serif;
font-size: 2.5rem;
font-weight: 700;
text-align: center;
text-transform: uppercase;
letter-spacing: 3px;
}

/* Event card description text */
.event-description {
font-size: 0.95rem;
font-style: italic;
text-align: justify;
line-height: 1.8;
letter-spacing: 0.5px;
color: #555;
}

/* Section sub-headings */
h2 {
font-family: 'Roboto', sans-serif;
font-size: 1.4rem;
font-weight: 700;
text-transform: capitalize;
letter-spacing: 1px;
}
```

```
}
```

[Screenshot – Browser showing Google Font applied + Network tab]

Browser screenshot shows:

- Network tab: fonts.googleapis.com 200 OK (font loaded)
- #mainHeader: Playfair Display, uppercase, letter-spacing 3px
- body text: Roboto regular, line-height 1.7
- .event-description: italic, justified, line-height 1.8
- h2: Roboto bold, capitalised

Exercise 6: Link and List Styling

Style anchor pseudo-classes and customise list bullets and spacing.

styles.css – link pseudo-classes

```
/* Unvisited link */
a:link {
  color: #1a73e8;
  text-decoration: none;
}

/* Visited link */
a:visited {
  color: #6a0dad;
}

/* Mouse hover */
a:hover {
  color: #0d47a1;
  text-decoration: underline;
}

/* Active (while clicking) */
a:active {
  color: #e74c3c;
}
```

styles.css – list styling

```
/* Event list – custom bullets */
.event-list {
  list-style-type: square;
  list-style-position: inside;
  padding: 0 0 0 16px;
  margin: 8px 0;
}

.event-list li {
  padding: 6px 0;
  margin-bottom: 4px;
  color: #444;
}

/* Navigation menu – no bullets */
nav ul {
  list-style-type: none;
  margin: 0;
  padding: 0;
  display: flex;
  gap: 20px;
}

nav ul li a {
  padding: 8px 14px;
  color: white;
```



```
font-weight: 600;
}

nav ul li a:hover {
background-color: rgba(255,255,255,0.2);
border-radius: 4px;
}
```

[Screenshot – Browser showing styled links and lists]

Browser screenshot shows:

- Unvisited links: blue #1a73e8, no underline
- Hovered link: darker blue + underline appears
- Visited link: purple #6a0dad
- Active link (while clicking): red flash
- .event-list: square bullets, items spaced 6px apart
- nav ul: horizontal flex menu, no bullets, links white

Exercise 7: Table Styling

Style the events admin table with borders, padding, zebra striping, and collapsed borders.

index.html – events admin table

```
<table class='events-table'>
<thead>
<tr>
<th>Event Name</th>
<th>Date</th>
<th>Location</th>
<th>Seats</th>
</tr>
</thead>
<tbody>
<tr><td>Summer Fair</td><td>12 Aug</td><td>Town Park</td><td>200</td></tr>
<tr><td>Book Club</td><td>19 Aug</td><td>Library</td><td>30</td></tr>
<tr><td>Coding Workshop</td><td>25 Aug</td><td>Tech Hub</td><td>50</td></tr>
<tr><td>Farmers Market</td><td>02 Sep</td><td>High Street</td><td>100</td></tr>
</tbody>
</table>
```

styles.css – table

```
/* Outer table */
.events-table {
width: 100%;
border-collapse: collapse;
margin: 20px 0;
font-size: 0.95rem;
}

/* Header cells */
.events-table th {
background-color: #2c3e50;
color: white;
padding: 12px 16px;
text-align: center;
border: 1px solid #2c3e50;
}

/* All data cells */
.events-table td {
padding: 10px 16px;
text-align: center;
border: 1px solid #dddddd;
color: #333;
}

/* Zebra striping on even rows */
.events-table tbody tr:nth-child(even) {
background-color: #f2f6fc;
}
```

```
/* Hover highlight */
.events-table tbody tr:hover {
background-color: #d0e4ff;
}
```

[Screenshot – Browser showing styled zebra-striped admin table]

Browser screenshot shows:

- Table header: dark background (#2c3e50), white text, centred
- All borders collapsed – no double-lines between cells
- Even rows (Book Club, Farmers Market): light blue #f2f6fc
- Odd rows: white background
- Hovered row: highlighted in #d0e4ff
- All text centred via text-align: center

Exercise 8: Box Model and Layout Control

Demonstrate margin, padding, border, outline on `.eventCard`, and compare `visibility:hidden` vs `display:none`.

styles.css – box model

```
/* Event card – box model properties */
.eventCard {
width: 100%;
padding: 20px; /* space inside border */
margin: 16px 0; /* space outside border */
border: 2px solid #1a73e8; /* visible border */
border-radius: 8px;
background-color: white;
box-sizing: border-box;
}

/* Outline on focused/selected form fields */
input:focus,
textarea:focus {
outline: 3px solid #1a73e8;
outline-offset: 2px;
}

/* visibility: hidden – element is hidden but STILL takes up space */
.hidden-but-space {
visibility: hidden;
}

/* display: none – element is removed from layout, takes NO space */
.removed-from-layout {
display: none;
}
```

[Screenshot – DevTools Box Model panel showing padding / border / margin]

```
DevTools > Elements > .eventCard – Computed panel shows:
content: (width minus padding)
padding: 20px on all sides
border: 2px solid blue
margin: 16px top and bottom

Form field when clicked: blue outline (3px) appears outside border.

visibility:hidden -> element invisible but gap remains in layout.
display:none -> element completely gone, layout collapses.
```

Note: outline is drawn outside the border and does not affect layout (it does not add to the element's box dimensions). Use it for focus indicators to meet accessibility requirements.

Exercise 9: Multiple Columns in Text

Display a community news article in a two-column newspaper layout.

index.html – news section

```
<section class='news-bulletin'>
<h2>Community Bulletin</h2>
<div class='news-content'>
<p>The local council has announced plans for a new community garden
on Elm Street. Residents are invited to join the planning meeting
on Saturday at 10am in the Town Hall.</p>
<p>The summer reading programme at the library has seen record
participation this year, with over 300 children signing up.
Final awards will be presented on 30 August.</p>
<p>Road maintenance work on High Street is scheduled for next
Monday. Drivers are advised to use alternative routes.
Work is expected to be completed within three days.</p>
</div>
</section>
```

styles.css – multi-column layout

```
/* Newspaper-style two-column layout */
.news-content {
column-count: 2;
column-gap: 30px;
column-rule: 1px solid gray;
text-align: justify;
font-size: 0.95rem;
line-height: 1.7;
}

/* Section heading above the columns */
.news-bulletin h2 {
column-span: all; /* heading spans across both columns */
text-align: center;
margin-bottom: 16px;
}
```

[Screenshot – Browser showing two-column news bulletin layout]

Browser screenshot shows:

- Text split evenly into 2 side-by-side columns
- 30px gap between the two columns
- A thin grey vertical line (column-rule) between them
- 'Community Bulletin' h2 spans full width above both columns
- Text is justified within each column

Similar to a printed newspaper layout.

Exercise 10: Responsive Web Design with Media Queries

Add a media query for screens below 768px, stack navigation vertically, and use flexible units. Bonus: Flexbox and Grid.

styles.css – base (desktop) layout

```
/* Desktop: horizontal nav */
nav ul {
  display: flex;
  flex-direction: row;
  gap: 20px;
  list-style: none;
  margin: 0;
  padding: 0;
}

/* Flexible image */
.banner-image {
  width: 100%;
  max-width: 800px;
  height: auto;
}

/* Events grid – desktop: 3 columns */
.events-grid {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  gap: 20px;
  width: 90%;
  margin: 0 auto;
}
```

styles.css – media query for tablets and phones (<768px)

```
/* ---- Mobile / Tablet breakpoint ---- */
@media screen and (max-width: 768px) {

  /* Stack nav links vertically */
  nav ul {
    flex-direction: column;
    gap: 8px;
    padding: 10px;
  }

  /* Reduce heading size */
  #mainHeader {
    font-size: 1.4rem;
    letter-spacing: 1px;
    padding: 14px 10px;
  }

  /* Single column grid on mobile */
  .events-grid {
    grid-template-columns: 1fr;
    width: 95%;
  }
}
```

```
/* Reduce image max-width */
.banner-image {
max-width: 100%;
}

/* Reduce body font size */
body {
font-size: 14px;
}

/* Stack columns into single column on mobile */
.news-content {
column-count: 1;
}
}
```

[Screenshot – DevTools Device Toolbar showing mobile layout]

DevTools > Toggle Device Toolbar > iPhone SE (375px wide):

- nav links stack vertically one per line
- font-size drops to 14px
- .events-grid shows 1 column instead of 3
- .news-content shows single column text
- banner-image fills full width

DevTools > Toggle Device Toolbar > iPad (768px wide):

- Media query not triggered (768px is the boundary)
- Nav still horizontal

Responsive breakpoint confirmed at exactly 768px.

Note: Use % and vw/vh for widths and heights that scale with screen size. Avoid fixed pixel widths on containers. Flexbox and CSS Grid together handle most responsive layout needs.

Exercise 11: Debug and Test with DevTools and VS Code

Use Chrome DevTools to inspect, test, and debug the portal layout.

Steps performed:

- Open Chrome. Press F12 to open DevTools.
- Click the device toolbar icon (Ctrl+Shift+M) to switch to responsive mode.
- Select preset devices: iPhone SE, iPad, Laptop 1366px.
- In the Elements panel, click an element to inspect its styles on the right.
- Uncheck / modify CSS values live in the Styles pane to test different values.
- Go to Network tab, reload page (F5), confirm styles.css returns 200 OK.

What to look for in DevTools

```
Elements panel > Styles pane:  
- Strikethrough rule = overridden by a higher-priority rule  
- Yellow triangle = invalid property value  
  
Computed pane:  
- Shows final resolved values (e.g. font-size: 16px from 1rem)  
- Box model diagram (margin / border / padding / content)  
  
Network tab:  
Name Status Type  
index.html 200 document  
styles.css 200 stylesheet <- external CSS loaded OK  
bg-pattern.png 200 image
```

[Screenshot – DevTools Network tab + Device toolbar + Elements panel]

```
DevTools – Elements panel:  
Clicking .eventCard highlights it in the viewport.  
Styles pane shows all applied rules in order of specificity.  
Overridden rules appear with strikethrough text.  
  
DevTools – Network tab after reload:  
styles.css 200 OK 1.2 kB  
(If 404 appeared here it would mean wrong file path in <link>)  
  
DevTools – Device toolbar at 375px (iPhone SE):  
Layout shifts to single-column as expected by media query.  
No horizontal scroll present – layout is fully responsive.  
  
VS Code – Prettier extension formats styles.css on save (Ctrl+S).
```

Note: Live edits in DevTools are temporary – they reset on page reload. Copy the tested values back into styles.css in VS Code to save them.

Summary – CSS Properties Used

Exercise	Key CSS Properties / Concepts
1 – CSS methods	inline style, <style>, <link>, /* comments */
2 – Syntax	Consistent indentation, /* section comments */
3 – Selectors	*, h2, #id, .class, h3 p (grouping)
4 – Colours	HEX, rgba(), background-image, linear-gradient()

5 – Typography	@import / <link> Google Font, font-family, font-size, font-weight, font-style, text-align, text-transform, letter-spacing, line-height
6 – Links & Lists	:link :visited :hover :active, list-style-type, list-style-position, display:flex, gap
7 – Tables	border-collapse:collapse, nth-child(even), th/td padding, text-align:center
8 – Box Model	padding, margin, border, outline, outline-offset, visibility:hidden, display:none, box-sizing
9 – Multi-column	column-count, column-gap, column-rule, column-span
10 – Responsive	@media, flex-direction:column, grid-template-columns:1fr, %, vw, vh, max-width:100%
11 – DevTools	Elements panel, Styles/Computed pane, Network tab, Device toolbar, live CSS editing